# Understanding Digital Data & Storage Systems

Edit with WPS Office

# What Is Digital Data?

**Digital data** is information represented as discrete binary values — 0s and 1s — rather than a continuously varying physical signal.

Every form of digital content — a document, a photo, a song, a video call — is ultimately a sequence of bits.

## Analog vs. Digital

### Analog

Continuous signal; infinite possible values (e.g., a vinyl record groove, a mercury thermometer)

### Digital

Discrete binary values; finite, precise states — easy to copy, compress, transmit, and error-check

Edit with WPS Office

# Types of Digital Data

*Most data generated today is unstructured — driving the shift toward flexible, distributed storage systems.*

S

## Structured

Organized into fixed rows/columns with a defined schema.

*Databases, spreadsheets*

S+

## Semi-structured

Has tags or markers for organization, but no rigid schema.

*JSON, XML, emails*

U

## Unstructured

No predefined format or organization.

*Images, video, audio, free text*

Edit with WPS Office

# What Is a Storage System?

**A storage system** is the combination of hardware, media, and software that records digital data and makes it available for later retrieval.

1

### Storage medium

The physical material data is written to — disk platters, flash cells, tape, optical discs.

2

### Controller / interface

Hardware and protocols (e.g., SATA, NVMe) that move data between the medium and the rest of the system.

3

### File system / software layer

Organizes raw storage into files and directories a user or program can address.

Edit with WPS Office

# Volatile vs. Non-Volatile Storage

*A key distinction in how storage systems retain data once power is removed.*

V

## Volatile Storage

Requires continuous power to retain data — contents are lost when power is switched off.

*Examples: RAM, CPU cache*

Used for fast, temporary working data during active processing.

NV

## Non-Volatile Storage

Retains data even when power is removed — persists until deliberately changed or erased.

*Examples: HDD, SSD, optical discs, tape*

Used for long-term storage of files, programs, and archives.

Edit with WPS Office

# Data Representation & Units

*Bit → Byte, then powers of 1024 (or 1000) as data scales up.*

| | |
|---|---|
| Bit | 1 binary digit (0 or 1) |
| Byte | 8 bits |
| KB | ~1,000 bytes |
| MB | ~1,000 KB |
| GB | ~1,000 MB |
| TB | ~1,000 GB |
| PB | ~1,000 TB |
| EB / ZB / YB | Data-center & global-scale volumes |



Edit with WPS Office

# The Storage Hierarchy

*A tradeoff between speed, cost per GB, and persistence.*

### Primary Storage

RAM, cache — fastest, volatile, most expensive per GB

### Secondary Storage

HDD, SSD — persistent, moderate speed and cost

### Tertiary Storage

Tape, optical archive — slow, cheapest, for backup/archival

Edit with WPS Office

# Storage Devices

HD

## HDD

*Magnetic*

Spinning platters, moving read/write head. Cheap per GB, mechanically slower, moving parts.

SS

## SSD

*Solid-State*

Flash memory, no moving parts. Much faster, pricier per GB, finite write cycles.

Op

## Optical

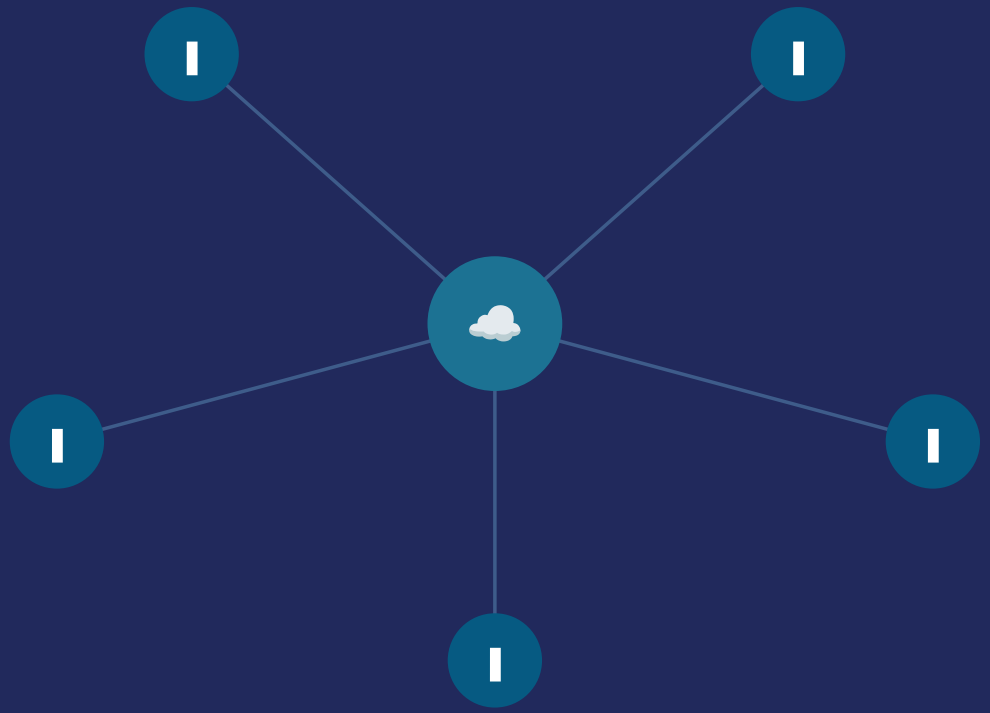
*CD/DVD/Blu-ray*

Laser-read discs. Mostly legacy today; still used for some archival cases.

Edit with WPS Office

# Cloud & Distributed Storage

*Modern systems spread data across many machines and locations — for scale and fault tolerance.*



- Object storage (e.g., cloud buckets) for massive, unstructured data
- Distributed file systems (e.g., HDFS) split data across clusters
- Replication across nodes/regions protects against hardware or site failure
- Enables elastic scaling — capacity grows with demand

Edit with WPS Office